

NIPUN ADARSH

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SUMMARY

Cyber Security student at DSCE with a track record of building security tools that work against real attack scenarios. Designed and shipped an ML-based intrusion detection framework and a decentralised threat coordination prototype from scratch, applying engineering principles throughout. Picks up new targets fast and works from first principles. Seeking a penetration testing internship to identify, assess, and help close security flaws across complex systems.

TECHNICAL SKILLS

Penetration Testing: Kali Linux, Burp Suite (web app testing), Nmap, Hping3

Network Analysis: Wireshark (packet capture & traffic analysis), Snort (IDS/IPS rules), Nmap (port & service scanning)

Programming: Python, Java, C, C++, Bash, SQL, HTML/CSS, Node.js

Security Concepts: Vulnerability Assessment, Threat Analysis, Intrusion Detection, Network Security, Cryptography, Incident Response

Environments: Kali Linux, VirtualBox/VMware, Docker, Git/GitHub, Flask, Scikit-learn

PROJECTS

Traffic Analysis & Node Categorisation — ML Cybersecurity Framework

Python, Scikit-learn, Wireshark, Network Analysis

- GitHub - [github.com/NipunAdarsh/Traffic-Analysis-and-Node-Categorization-Machine-Learning-Integrated-Framework](#)
- Built a 3-model ML pipeline (Random Forest + Isolation Forest) identifying security vulnerabilities across 6 protocol types and 5 node categories; achieved 92% classification accuracy and 95% AUC, mapped to real-world attack patterns including Mirai Botnet and DDoS vectors
- Assessed vulnerability impact at 89% precision and 94% recall across simulated attack conditions; defined detection thresholds and recommended packet-level controls for anomalous traffic (5% injection rate, 40–2000 byte range)

SwarmNet — Decentralised Network Security Prototype

- GitHub - [github.com/NipunAdarsh/SwarmNet](#).
- Built a 5-node decentralised threat coordination system (4 NPU + 1 CPU) using UDP multicast discovery and TCP heartbeat, cutting simulated inference latency from 18–35ms (CPU) to 6–14ms (NPU) with automatic failover on 3 missed heartbeats
- Designed queue-aware load balancing across 4 protocols applying engineering principles drawn from distributed threat intelligence systems; 2-second heartbeat intervals with sub-6-second failover recovery

Digital Footprint Guardian — Cybersecurity Intelligence Dashboard

React 18, Supabase, Groq (Llama 3.3 70B), Python, Tailwind CSS

- GitHub - [github.com/NipunAdarsh/DIGITAL-FOOTPRINT-GUARDIAN](#)
- Built a full-stack threat intelligence platform with 6 security modules: email breach detection (Leak-Lookup API + SHA-256 privacy), phishing URL analysis using Levenshtein typosquatting against 16 major domains, and Shannon-entropy password strength scoring with crack-time estimation
- Integrated Groq (Llama 3.3 70B) for AI-driven remediation plans across breach, password, and phishing analysis; routed sensitive API keys server-side via Supabase Edge Functions to eliminate client-side key exposure

HACKATHONS & ACTIVITIES

Startup Hackathon — The Idea Company

2025

1st Runner-Up | Idea & Product Lead

- Led product strategy for a Python/Flask Resume Optimiser with ATS scoring and AI-powered job matching; placed 1st Runner-Up out of ~25 teams

EDUCATION

Dayananda Sagar College of Engineering, Bengaluru

Sep 2025 – 2028 (Expected)

B.E. Computer Science – Cyber Security Specialisation

- Coursework: Network Security, Cryptography, Vulnerability Assessment & Penetration Testing, Ethical Hacking, Digital Forensics, Cyber Risk Management, Secure Software Development, Cyber Law & Compliance, AI & Machine Learning

The Oxford Polytechnic, Bengaluru

Sep 2022 – May 2025

Diploma in Computer Science | GPA: 8.97 / 10 — Distinction

- Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management, Object-Oriented Programming, Linux Administration

CERTIFICATIONS

Cisco Networking Academy — Introduction to Cybersecurity — Completed Mar 2026.

Tata Group (Forage) — Cybersecurity Analyst Job Simulation — Completed Apr 2026.

Anthropic — Claude 101 — Completed Jun 2026.

Anthropic — Claude Code 101 — Completed Jun 2026.